**KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY**
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KALINGA INSTITUTE OF
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Submitted in partial fulfillment of the requirements for the degree of
**Bachelor of Technology**
in
**Computer Science & Engineering**


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Kalinga Institute of Industrial Technology (KIIT)

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# 1. Introduction

## 1.1 Purpose

ProxiChat is a web app built exclusively for close range peer to peer communication. It aims to use Bluetooth technology to connect to peers located in close proximity and use to chat in conditions where internet is unavailable or jammed. Unlike previous systems where Bluetooth is used for only device detection, here messages are sent using Bluetooth Connections removing even more dependency from traditional Internet services, hence providing a low latency experience for users.

## 1.2 Product Scope

ProxiChat serves as an alternative to traditional communication application requiring internet connection for communication. Its scope includes:

1. Device Detection: Quick Device Detection using Bluetooth MAC Address of login device.
2. Connection Management: Allow Users to select a nearby device and connect to them while also performing handshaking and authentication before communication starts.
3. Chat Functionality: Exchange chat messages in real time over Radio Frequency Communication.

The handshaking and authentication before communication between two users ensures that the chat is completely end to end encrypted without using internet for transfer data between devices, along with data being stored locally in SQLite provides another level of protection for user data.

## 1.3 Definitions, Acronyms, and Abbreviations

- SRS: Software Requirement Specification
- UI: User Interface
- UX: User Experience
- API: Application Programming Interface
- IDE: Integrated Development Environment
- DB: Database
- SQL: Structured Query Language
- HTTP: Hypertext Transfer Protocol
- GUI: Graphical User Interface
- BE: Base band Entity
- BLE: Bluetooth Low Energy
- RFCOMM: Radio Frequency Communication
- JSON: JavaScript Object Notation

## 1.4 Technologies to be used

1. Frontend: React, TypeScript, vite, Tailwind CSS, Lucide React, WPF, XAML.
2. Backend: npm, ESLint, C#(.NET 8).
3. Database: SQLite.
4. Development and Build Tools: Visual Studio, Visual Studio Code, .NET SDK, NuGet Packages, dotnet, GitHub, NPM.

# 2. Overall Description

## 2.1 Product Perspective

ProxiChat is a web-app that integrates Bluetooth connectivity for effective and end to end safe communication between users. This helps use an alternative source of networks for communication between users when traditional internet systems do not operate by establishing a peer to peer network.

## 2.2 User Classes and Characteristics

1. End Users (Primary): The primary users of ProxiChat are individuals who require offline, peer-to-peer communication in environments where internet connectivity is unavailable or unreliable. This includes students, travelers, and emergency responders. For non-technical users with basic computer skills, expecting a simple and intuitive user interface.
2. Admins/ Event Organizers (Future): Admins may moderate message content, view analytics, and configure device discovery parameters.

## 2.3 Operational Parameters

1. Windows Platform:
   a) Supported on devices running Windows 11, ensuring compatibility between devices.
2. Connectivity:
   a) Requires an active Bluetooth connection which is required for connection between users.

## 2.4 Software Interfaces

1. Frontend: This web application is built using React and Tailwind CSS compatible with Windows 11.
2. Bluetooth Communication: This app uses Bluetooth stacks for discovery, pairing and messaging.
3. Database: Local SQLite databases store user profile, chat history and device connections.

## 2.5 Hardware Interfaces

**1.** This application requires standard computers with Bluetooth Hardware. No additional hardware is required.

## 2.6 External Forces

### 2.6.1 Dependancies

1. Device must support Bluetooth (classic or BLE) and have sufficient battery capacity.

2. SQLite for local data persistence.

3. React for platform UI.

### 2.6.2 Constraints

1. Technology Constraint: App functionality limited to Bluetooth only devices.

2. Security Constraint: Messages must be encrypted (AES-256 or similar) to prevent eavesdropping.

3. Database Constraint: Chat history is stored locally, Cloud storage is optional.

4. Platform Constraint: Limited by Bluetooth range (typically 10-30 meters).

5. Scope Constraint: Group chat and admin moderation features are planned for future development.

# 3.System Features

## 3.1 Functional Requirements

### FR1 User Account Creation:

**FR 1.1:**

-Input: User selects Create Account.

-Output: Access granted to Create Account Page.

-Processing: Provide access to Create Account Page.

**FR 1.2:**

-Input: Enter valid Username and Password.

-Output: Returned Access to Login Page.

-Processing: Stores Username, Password and Current MAC Address to SQLite.

**FR 1.3:**

-Input: Invalid Username and Password.

-Output: Create Account Error Displayed.

-Processing: Account Creation Fails, User stays on Create Account page.

### FR2 User Login:

**FR 2.1:**

-Input: Enter valid Username and Password.

-Output: Access granted to Main App.

-Processing: Redirected data to SQLite for data checking comes back as true.

**FR 2.2:**

-Input: Invalid Username and Password.

-Output: Stay in Login. Error message displayed.

-Processing: Redirected data to SQLite for data checking comes back as false.

**FR3 Navigation:**

**FR 3.1:**

-Input: User selects Add Device.

-Output: Access granted to Discover New Devices page.

-Processing: Opens Discover New Devices page and starts searching for devices running the application in the vicinity.

**FR 3.2:**

-Input: User selects Device to Communicate from Discover New Devices.

-Output: Fresh Chat Screen Opens.

-Processing: Completes authentication and handshaking with selected device and loads chat.

**FR 3.3:**

-Input: User selects Device from Sidebar.

-Output: Chat Screen Opens.

-Processing: Opens Chat Screen and allows user to send messages.

## FR 4: Communication

**FR 4.1:**

-Input: User enters message and sends.

-Output: Message is sent and displayed on chat window.

-Processing: Message data is sent via Bluetooth packets and message information is stored locally on SQLite.

## 3.2 Non-Functional Requirements

**1. Performance:**

Quick, efficient device discovery and low latency communication message delivery.

**2. Security:**

Messages will be encrypted using a standard encryption model and only paired devices will be allowed to communicate.

**3. Usability and User Experience:**

Application will have a simple UI with clear visual feedback on message sending, connection feedback and errors.

# 4. Design

## 4.1 System Design

Users interacting with login to access discovering nearby users, initiating chats and also sending and receiving messages.

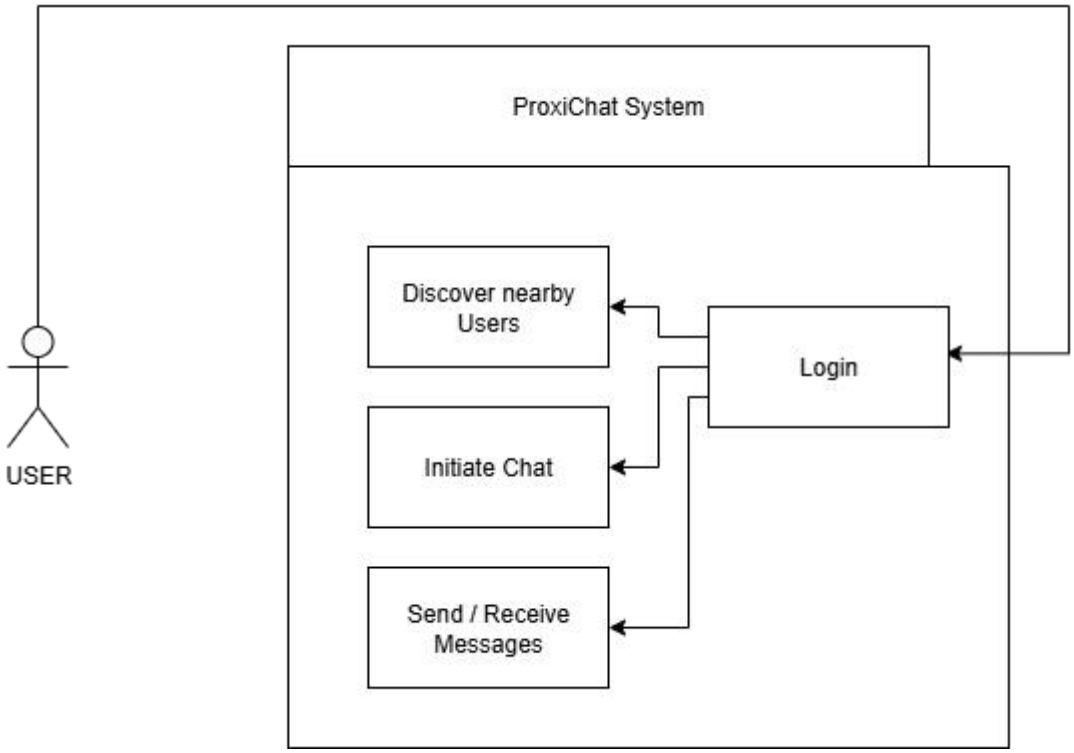



Fig 4.1: ProxiChat Use Case Diagram

## 4.2 Class Diagram

Core classes like User, Device, Bluetooth, Message, Message Handler and Database.

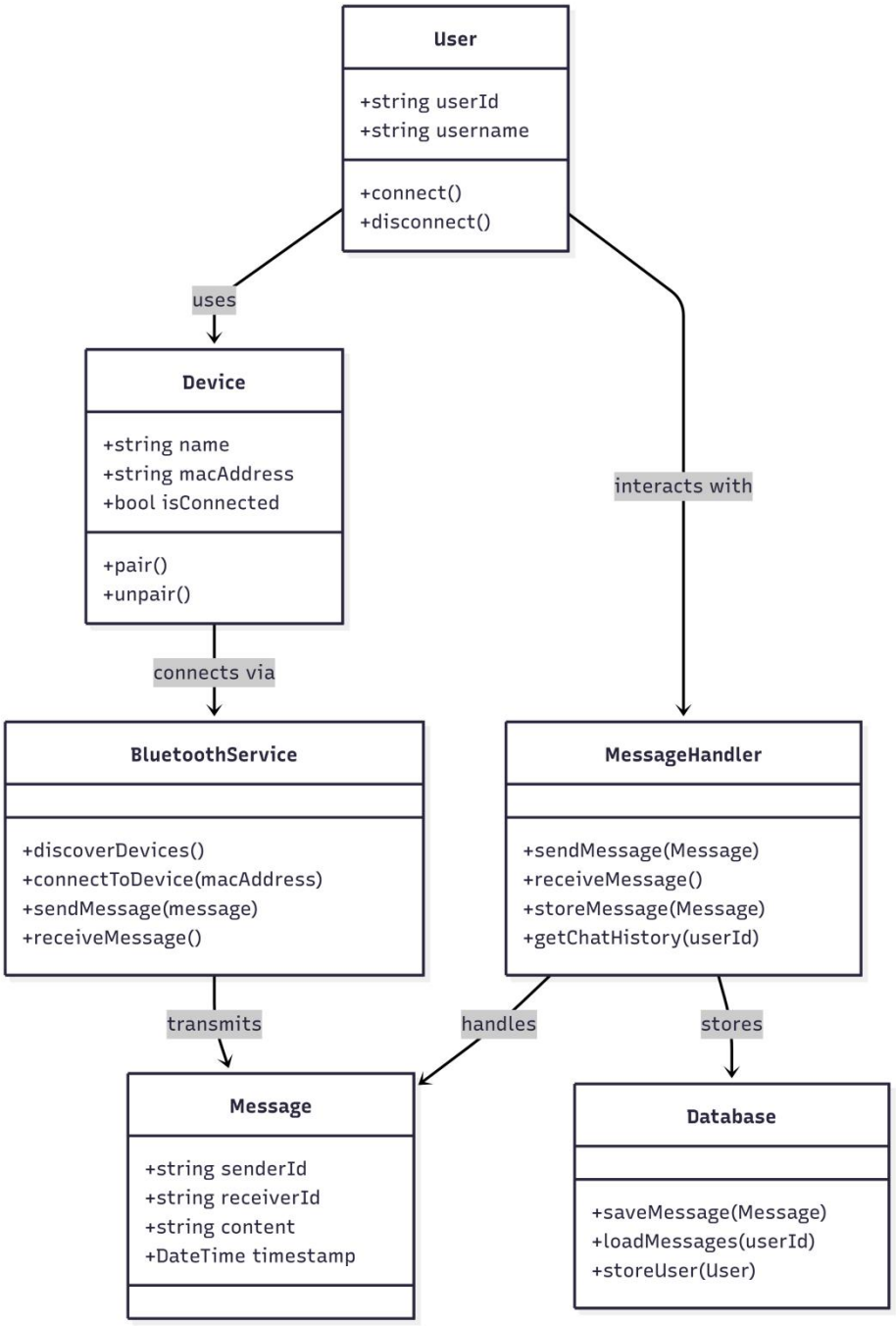



Fig 4.2: ProxiChat High Level Class Diagram

## 4.3 Sequence Diagram

Example: User A logging in and messaging User B.

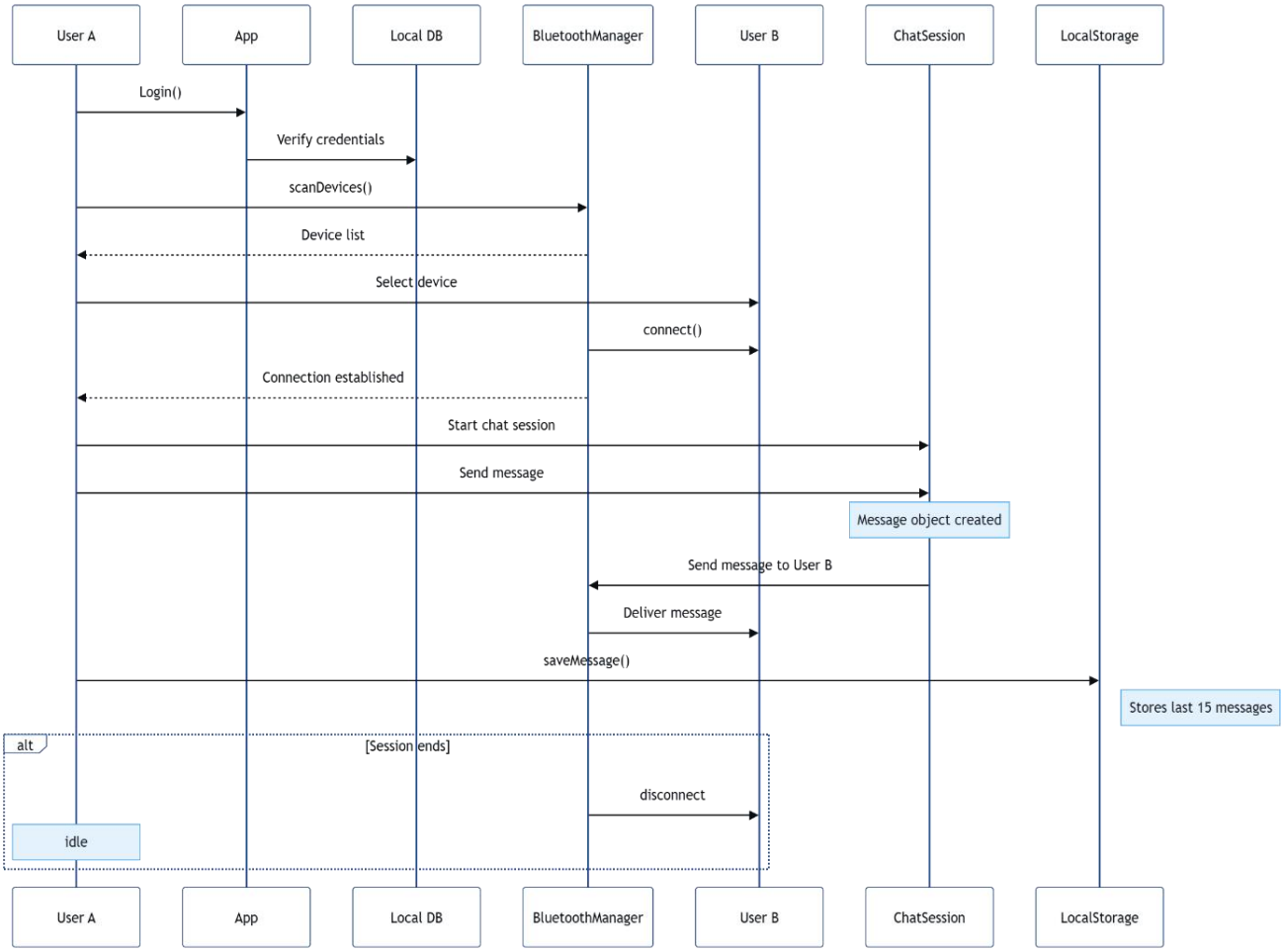



Fig 4.3: Sequence diagram for User A messaging User B

## 4.4 State Chart Diagram

Models the lifecycle of a chat session: Idle → Scanning → Connecting → Connected → Messaging

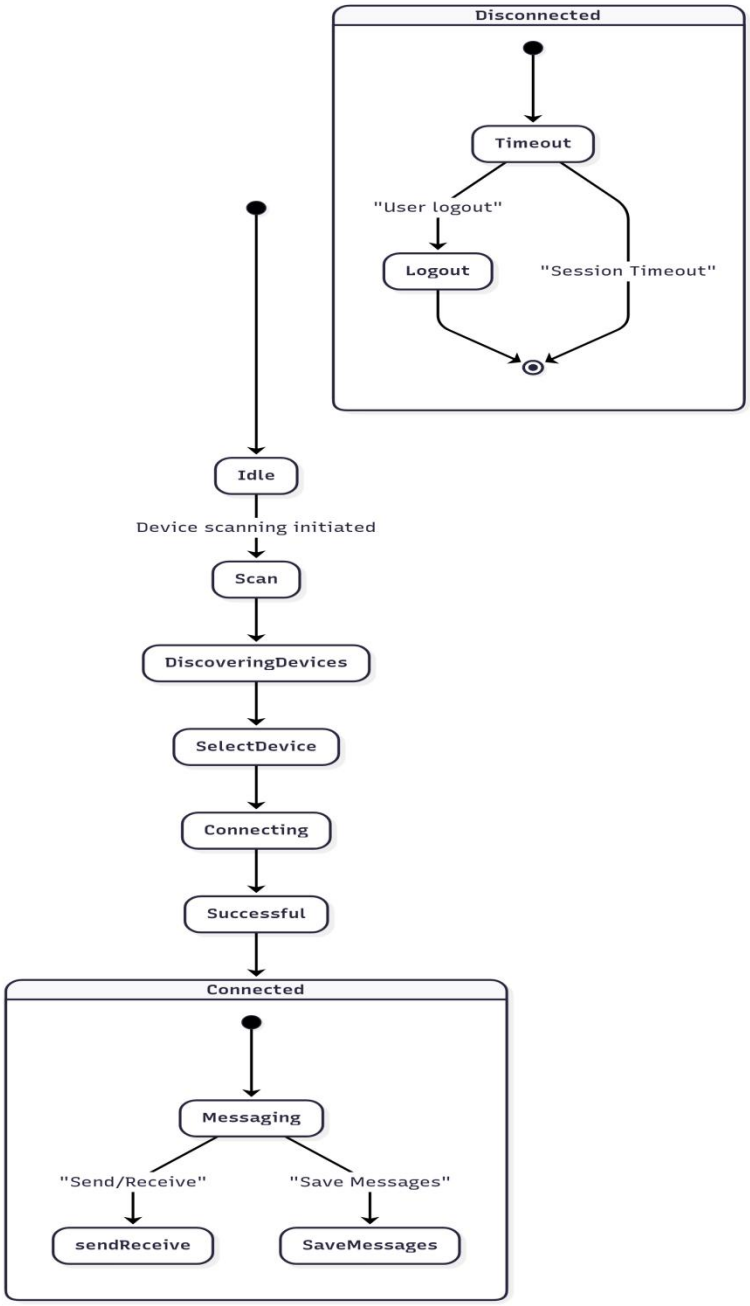



Fig 4.4 State Chart Diagram for Life Cycle of User

## 4.5 Activity Diagram

This diagram provides a visual flowchart of a process from start to finish. The following diagram illustrates the step-by-step activities involved when a sender do an message. Step-by-step flow for device discovery, pairing, and messaging process.

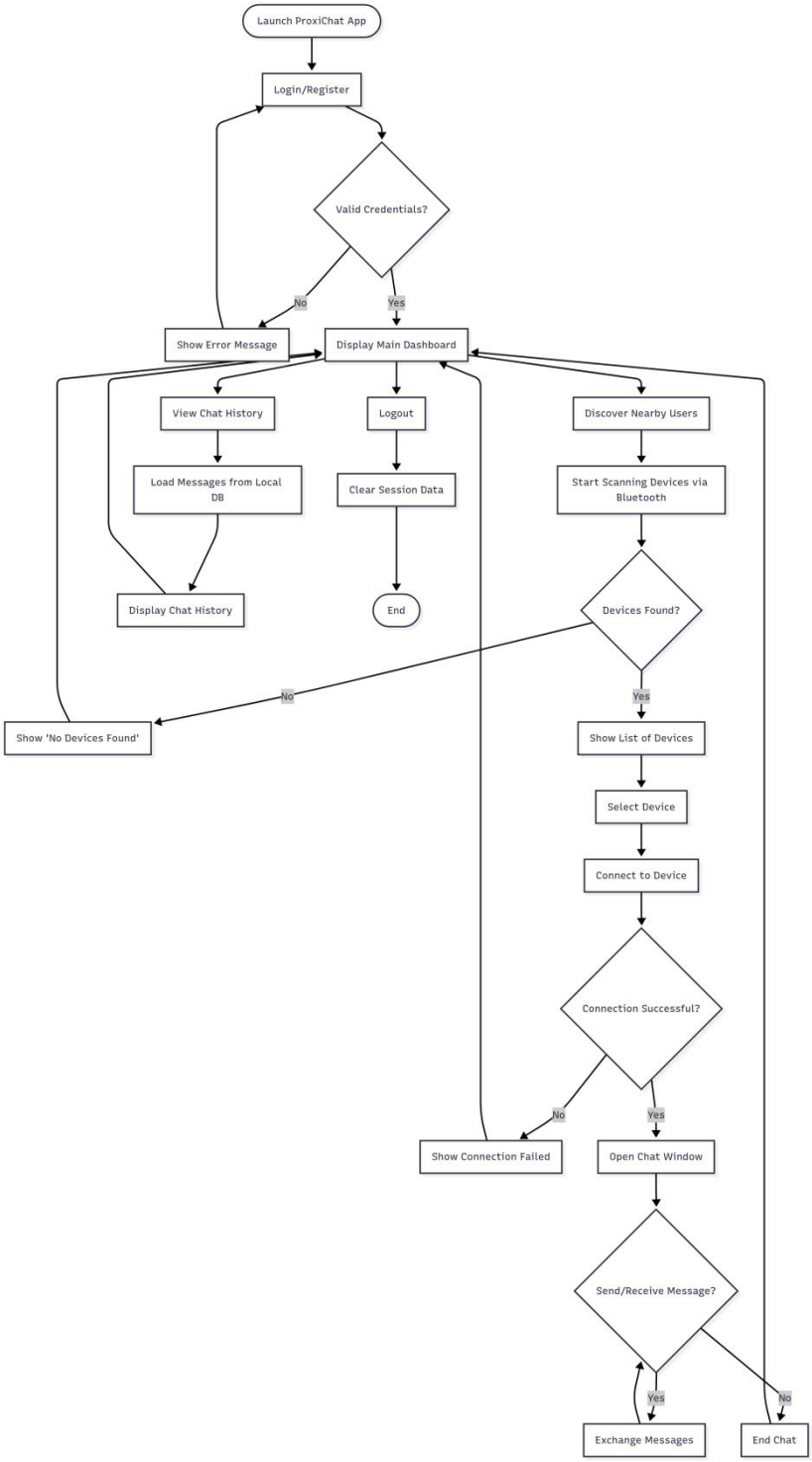



Fig 4.5: Activity Diagram for core ProxiChat functions

## 4.6 Data Flow Diagram

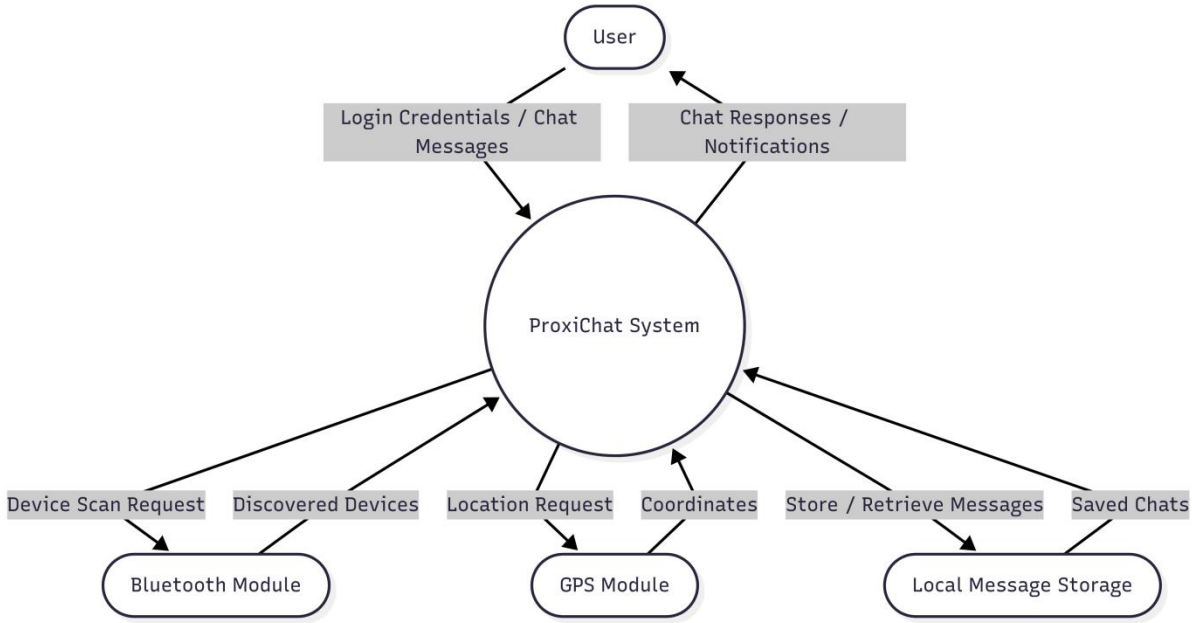



Fig 4.6.1: Level 0 Data Flow Diagram

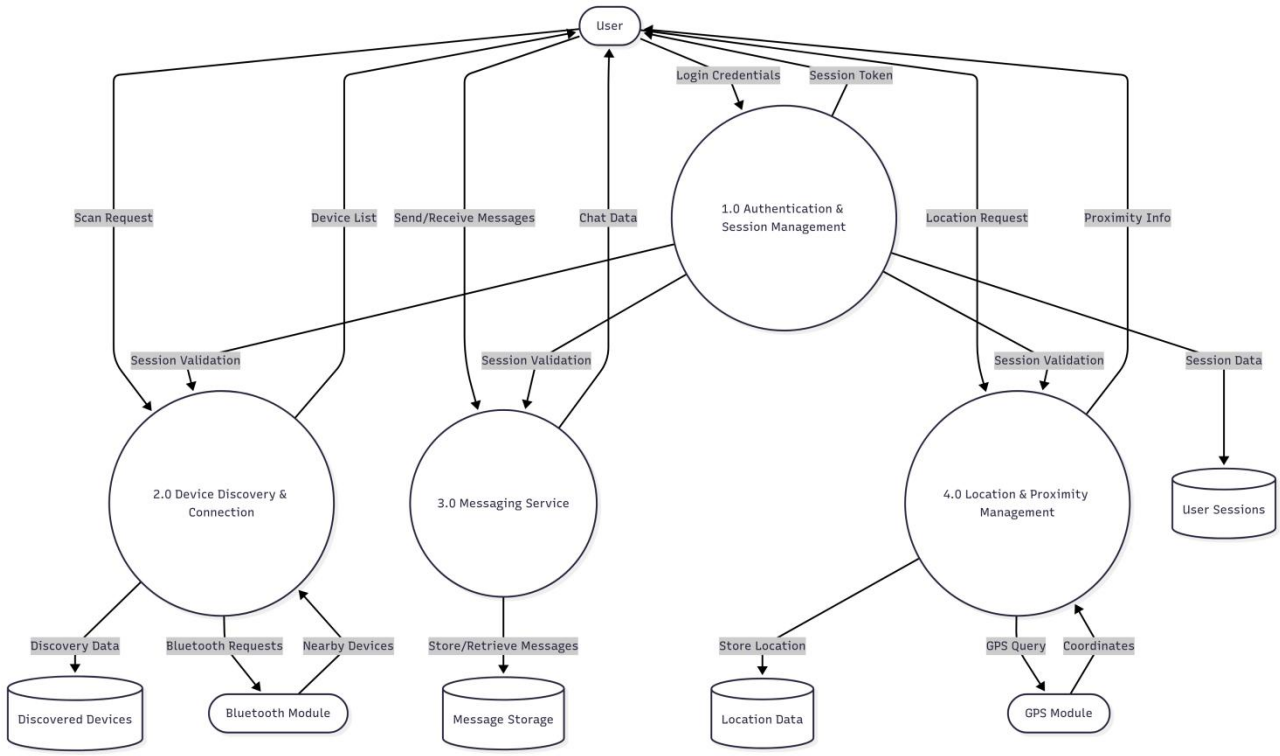



Fig 4.6.2: Level 1 Data Flow Diagram

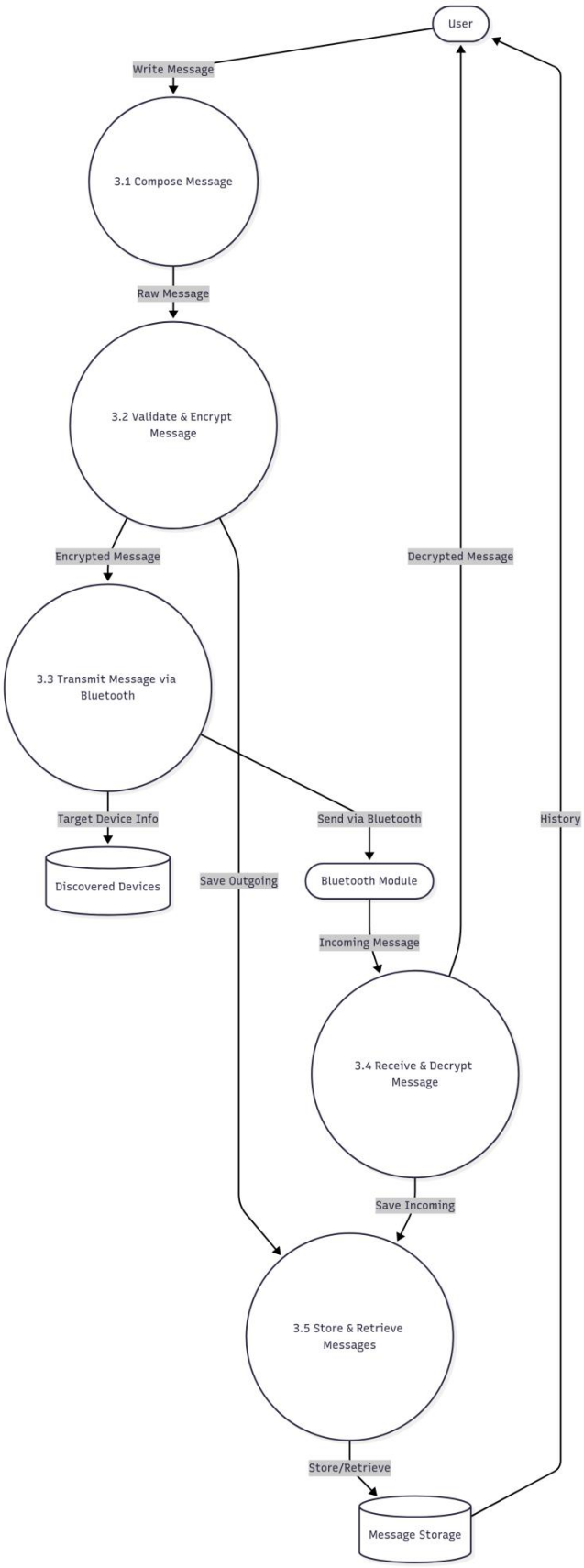
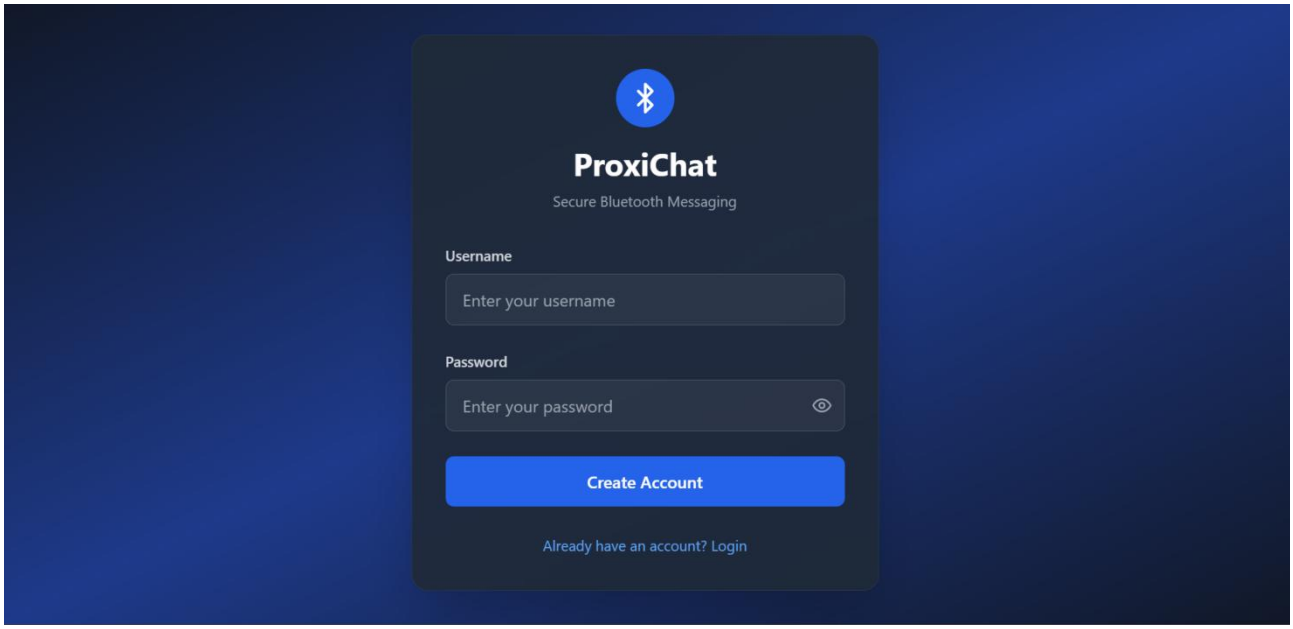
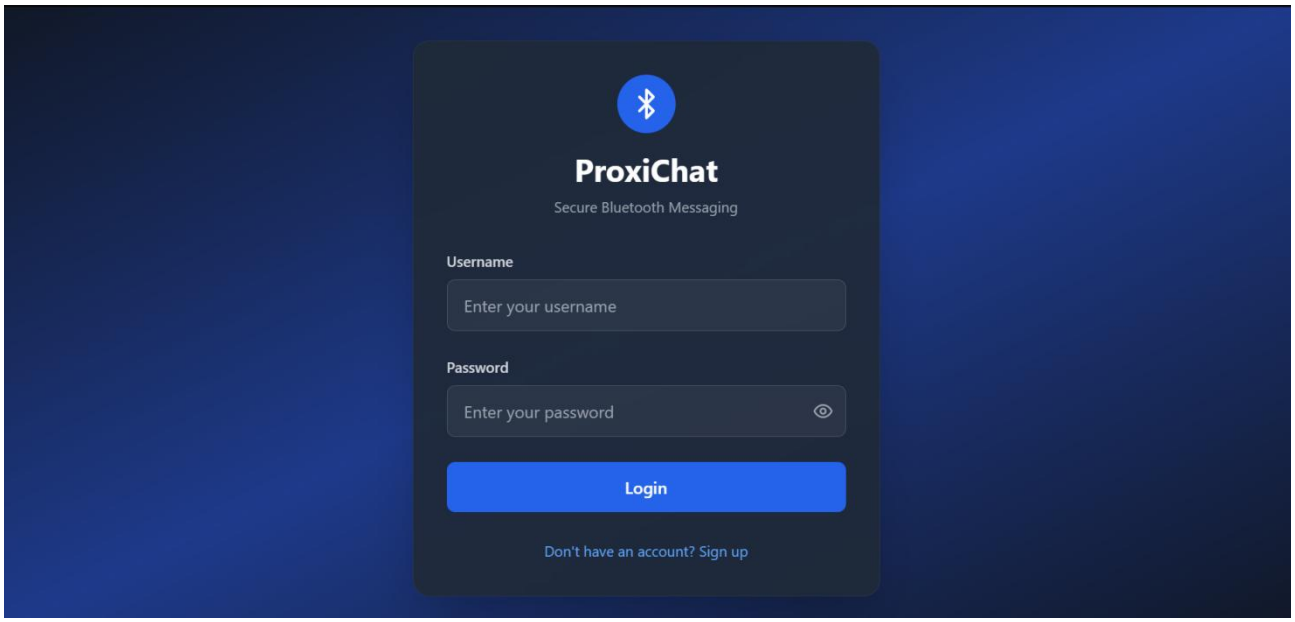
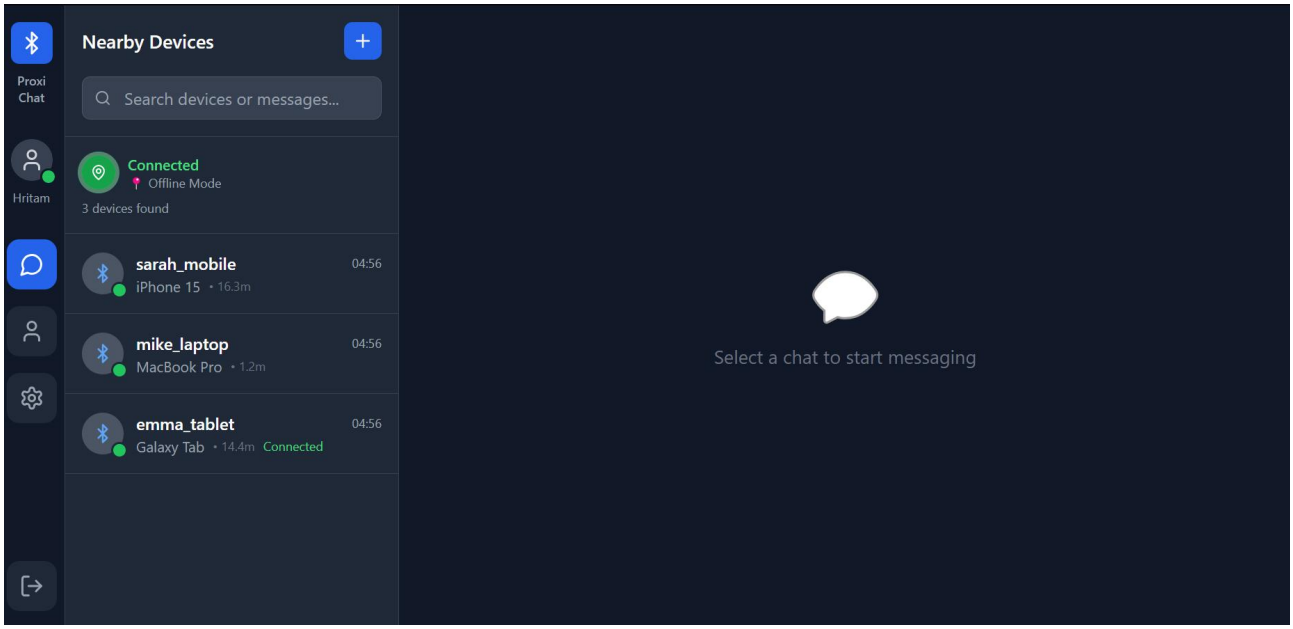


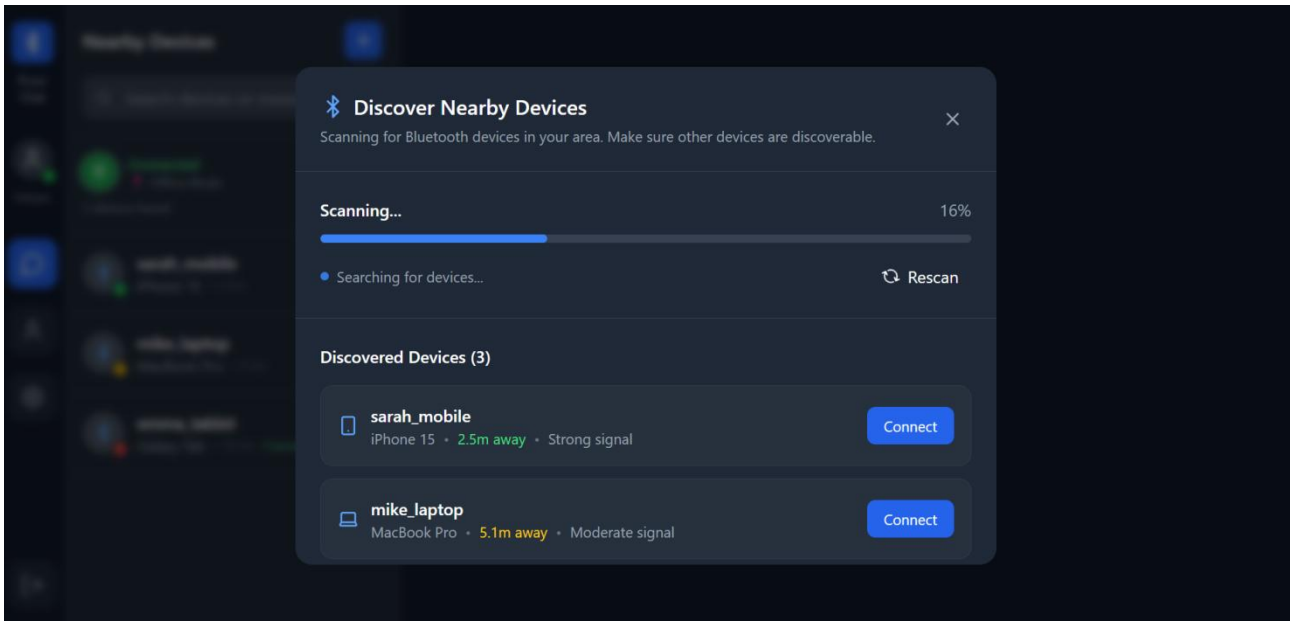

Fig 4.6.3: Level 2 Data Flow Diagram

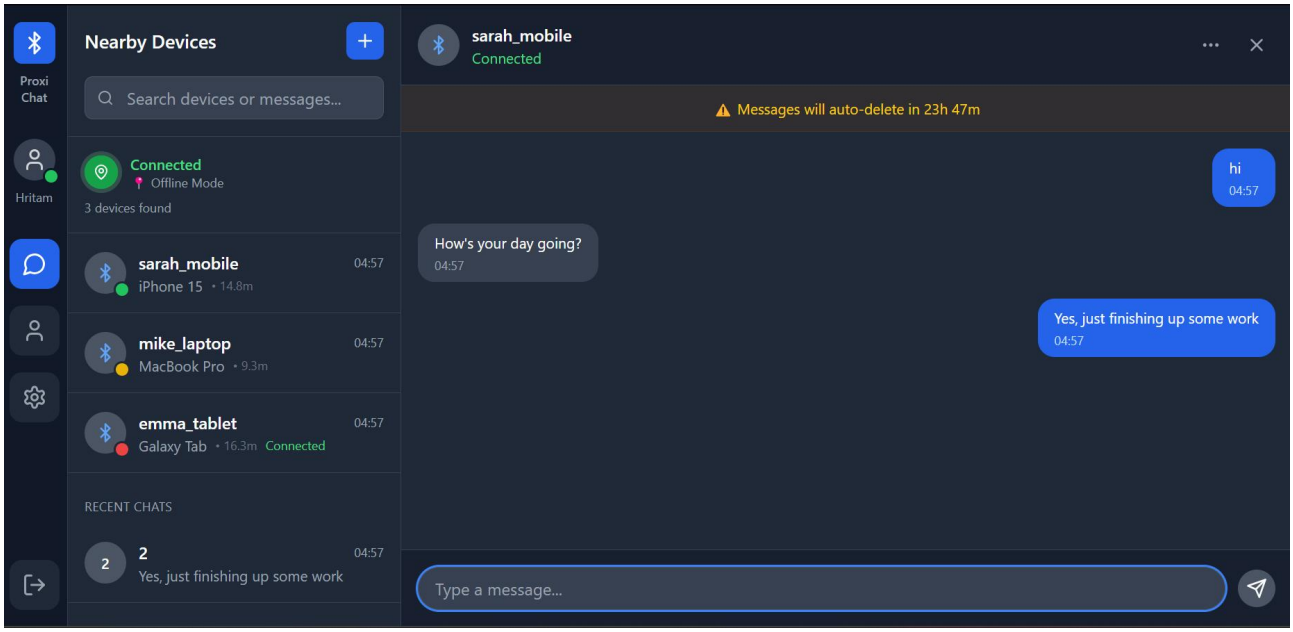
# 5. Graphical User Interface (GUI)




Proxi Chat
Hritam
Nearby Devices
Search devices or messages...
Connected
Offline Mode
3 devices found
sarah_mobile
iPhone 15 • 16.3m
04:56
mike_laptop
MacBook Pro • 1.2m
04:56
emma_tablet
Galaxy Tab • 14.4m Connected
04:56
Select a chat to start messaging


Discover Nearby Devices
Scanning for Bluetooth devices in your area. Make sure other devices are discoverable.
Scanning...
16%
Searching for devices...
Rescan
Discovered Devices (3)
sarah_mobile
iPhone 15 • 2.5m away • Strong signal
Connect
mike_laptop
MacBook Pro • 5.1m away • Moderate signal
Connect


Proxi Chat
Hritam
Nearby Devices
Search devices or messages...
Connected
Offline Mode
3 devices found
sarah_mobile
iPhone 15 • 14.8m
04:57
mike_laptop
MacBook Pro • 9.3m
04:57
emma_tablet
Galaxy Tab • 16.3m Connected
04:57
RECENT CHATS
2
Yes, just finishing up some work
04:57
sarah_mobile
Connected
Messages will auto-delete in 23h 47m
hi
04:57
How's your day going?
04:57
Yes, just finishing up some work
04:57
Type a message...

## 6. Future Scope

- Group Chat: Support Multiple devices in a single session.
- Cloud Backup: Store encrypted chat history in cloud for multi device access.
- File Sharing: Enable larger file and media attachment.
- Proximity Alerts: Notifications when new devices enter.
- Admin/Event Mode: Monitor or moderate multiple connections in organized events.

## 7. Conclusion

The Bluetooth Proximity Chat App aims to deliver a modern, user-friendly, and secure messaging solution for devices within Bluetooth range. It provides core functionalities for peer-to-peer communication, real-time chat, and local message persistence, all while maintaining smooth performance and an appealing interface. The architecture and design choices provide a strong foundation for future enhancements, including group chat, cloud integration, and advanced device management features.

## 8. References

Version Control: github.com
Dotnet (v9.0): https://dotnet.microsoft.com
Core Bluetooth (InTheHand.net): https://www.nuget.com
SQLite Documentation: https://www.sqlite.org/docs.html
Bluetooth RFCOMM: https://www.bluetooth.com
Google Fonts: https://fonts.google.com